

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

On

SmartLearn (An AI-Based Academic Support Platform)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **SmartLearn (AI-Based Academic Support Platform)** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Preeti Saini**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura
Date: 28-04-2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Students today widely use digital learning materials such as PDF notes, lecture slides, and online textbooks. Although these resources provide easy access to information, reading long documents can be time-consuming and it often becomes difficult for students to quickly identify important concepts or revise topics effectively before examinations. In many cases, students also rely on different tools for reading documents, generating summaries, and practicing quizzes. This separation of tools makes the learning process slow, unorganized, and less efficient.

The objective of this project is to develop an AI-based academic support system called **SmartLearn**, which helps students study more efficiently from PDF documents. The system integrates several learning tools into a single platform so that students can read documents, generate summaries, ask questions, create flashcards, and practice quizzes without switching between multiple applications. By combining these features, the platform aims to make digital learning more structured and convenient.

SmartLearn is developed as a web application using the **MERN stack**, which includes **MongoDB, Express.js, React.js, and Node.js**. The frontend interface is built with React.js and styled using Tailwind CSS to provide a responsive and user-friendly design. The backend is implemented using Node.js and Express.js to handle API requests and manage system operations, while MongoDB is used for storing user information, uploaded documents, flashcards, and quiz data. The system integrates Google Gemini AI to analyze uploaded documents and provide intelligent features such as document summarization, question answering, concept explanation, flashcard generation, and quiz generation.

After uploading a PDF document, users can read the document directly within the platform using an embedded PDF viewer. The AI system analyzes the document content and allows users to generate summaries, ask questions, and create flashcards or quizzes automatically. A progress dashboard helps users track their learning activities and monitor their study progress. Overall, the SmartLearn platform is designed to improve learning efficiency by reducing the time required to read lengthy documents and by providing effective revision tools such as summaries, flashcards, and quizzes. By integrating multiple study features into one system, SmartLearn helps students understand concepts more quickly and prepare for examinations in a more organized and productive manner.

1. Introduction to Project

1.1 Project Background

In recent years, digital learning resources have become an essential part of modern education. Students widely use PDF notes, lecture slides, research papers, and digital textbooks as their primary study materials. These digital resources provide flexibility, easy access, and the ability to store large amounts of academic information. However, as the amount of digital content increases, students often face difficulties managing and understanding lengthy documents efficiently.

Reading long PDF files can be time-consuming, especially when students need to quickly identify important concepts or prepare for examinations. During revision, students often struggle because they must repeatedly search through large documents to find key points. Additionally, students usually rely on multiple tools for different tasks such as reading documents, generating summaries, creating flashcards, or practicing quizzes. Using multiple applications makes the learning process fragmented, slow, and less organized.

With the rapid development of Artificial Intelligence (AI) technologies, new opportunities have emerged to improve digital learning systems. AI can analyze large amounts of text data, extract important information, and provide intelligent assistance to users. Integrating AI with learning platforms can significantly improve the way students interact with their study materials.

1.2 Problem Statement

Despite the availability of digital learning resources, students face several challenges when studying from PDF documents. One of the main issues is that reading and understanding lengthy documents requires considerable time and effort. Important concepts are often buried within large amounts of text, making it difficult for students to quickly locate relevant information.

Another challenge is the lack of efficient revision tools. Students usually do not have ready access to summarized notes, flashcards, or practice quizzes generated directly from their study materials. As a result, they spend additional time manually creating revision materials.

Furthermore, most existing learning tools provide only limited functionality. For example, one application may allow document reading, while another provides summarization or quiz generation. Switching between multiple tools disrupts the study process and reduces productivity. These limitations highlight the need for a unified system that integrates document reading, AI assistance, and revision tools in a single platform.

1.3 Objective of the Project

The main objective of the **SmartLearn** project is to develop an AI-powered academic support platform that helps students learn from PDF documents more efficiently and in a more structured manner.

The specific objectives of the project are:

- To allow users to upload and manage PDF study materials within a single platform.
- To integrate artificial intelligence for analyzing document content.
- To provide AI-based question answering from uploaded documents.
- To automatically generate concise summaries of long documents.
- To create flashcards for quick and effective revision.
- To generate multiple-choice quizzes for knowledge assessment.
- To provide a dashboard for tracking user learning activity and progress.

By achieving these objectives, SmartLearn aims to simplify digital learning and improve academic productivity for students.

1.4 System Overview

SmartLearn is designed as a web-based learning platform that combines document management with artificial intelligence to create an interactive academic environment. The system is implemented using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. Secure user authentication is implemented using JSON Web Tokens (JWT) to ensure safe access to the platform.

After logging into the system, users can upload their PDF study materials. The platform includes an embedded PDF viewer, allowing users to read documents directly within the application. The uploaded documents are processed and analyzed using Google Gemini AI, which enables several intelligent features.

Users can interact with their documents by asking questions, generating complete document summaries, or requesting explanations for specific concepts. In addition, the system automatically generates flashcards and multiple-choice quizzes from document content, helping students revise and test their knowledge efficiently.

The platform also includes a progress tracking dashboard that displays learning activity such as uploaded documents, generated flashcards, and quiz performance. Additional features such as favorites allow users to save important flashcards for quick review.

1.5 Expected Benefits

The SmartLearn platform is designed to significantly improve the digital learning experience for students. By integrating multiple study tools within a single system, the platform reduces the need for manual effort and eliminates the need to switch between different applications.

The system helps students save time when studying long documents, improves understanding through AI-based explanations, and supports faster revision using summaries, flashcards, and quizzes. The progress dashboard also helps students monitor their learning activities and stay organized during exam preparation.

Overall, SmartLearn provides a modern AI-driven approach to academic learning, making the study process more efficient, structured, and productive.

2. Methodology

The SmartLearn system is designed using a modular architecture where different components work together to provide intelligent learning assistance. This architecture divides the system into multiple modules such as authentication, document management, AI processing, flashcard generation, and quiz generation. Each module performs a specific task, which helps in improving system efficiency, scalability, and maintainability.

The modular design also makes it easier to update or improve individual components without affecting the entire system. This approach ensures that the SmartLearn platform operates in a structured and organized manner while providing reliable AI-powered learning features to users.

2.1 System Approach

The system follows a web-based architecture where the frontend interacts with the backend server through APIs. The backend processes requests, manages user data, and communicates with the AI service.

The major modules of the system include:

- User Authentication Module
- Document Upload and Management Module

- AI Processing Module
- Flashcard Generation Module
- Quiz Generation Module
- Dashboard and Progress Tracking Module

2.2 Tools and Technologies

The following technologies were used to develop the SmartLearn platform.

- **Frontend:** React.js is used to build a dynamic and responsive user interface.
- **Backend:** Node.js and Express.js are used to handle server operations and API development.
- **Database:** MongoDB is used to store user data, uploaded documents, flashcards, and quiz results.
- **Artificial Intelligence:** Google Gemini AI is used to analyze document content and generate summaries, explanations, flashcards, and quizzes.
- **User Interface:** Tailwind CSS is used to design a modern and responsive interface.
- **Development Environment:** Visual Studio Code is used as the primary development tool.

2.3 Flow of the System

The workflow of the SmartLearn system is as follows:

1. User registers and logs into the system.
2. The user uploads a PDF study document.
3. The system extracts text from the uploaded document.
4. Google Gemini AI processes the document content.
5. The system generates summaries, flashcards, quizzes, and answers questions.
6. The user can review results and track progress through the dashboard.

3. Design

3.1) Use Case Diagram:

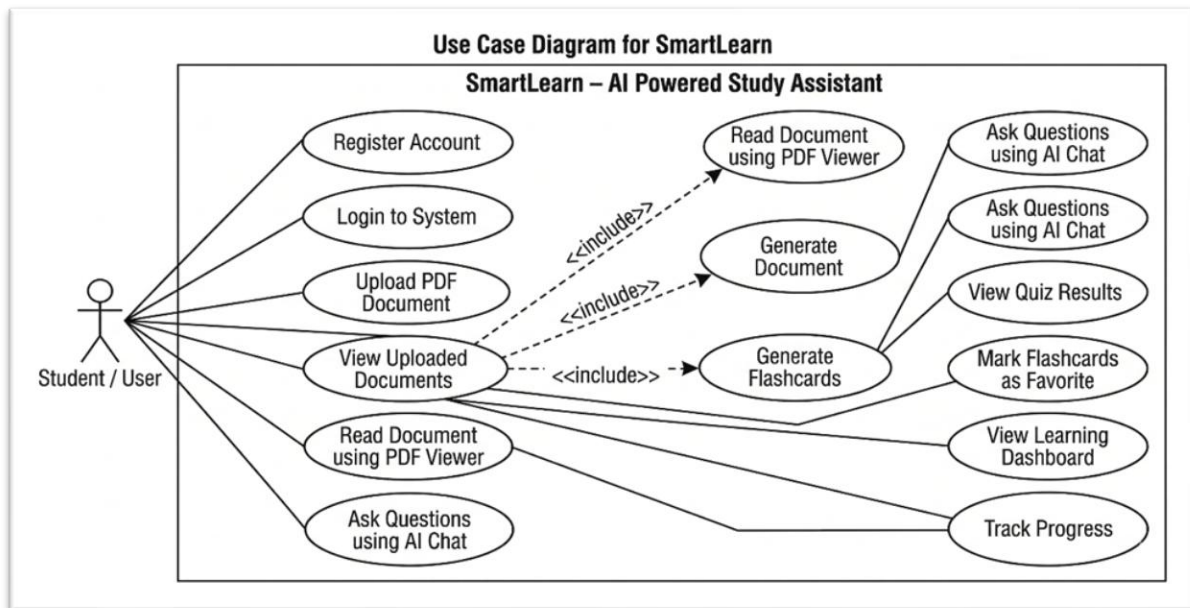


Fig: 3.1

3.2) System Architecture Diagram:

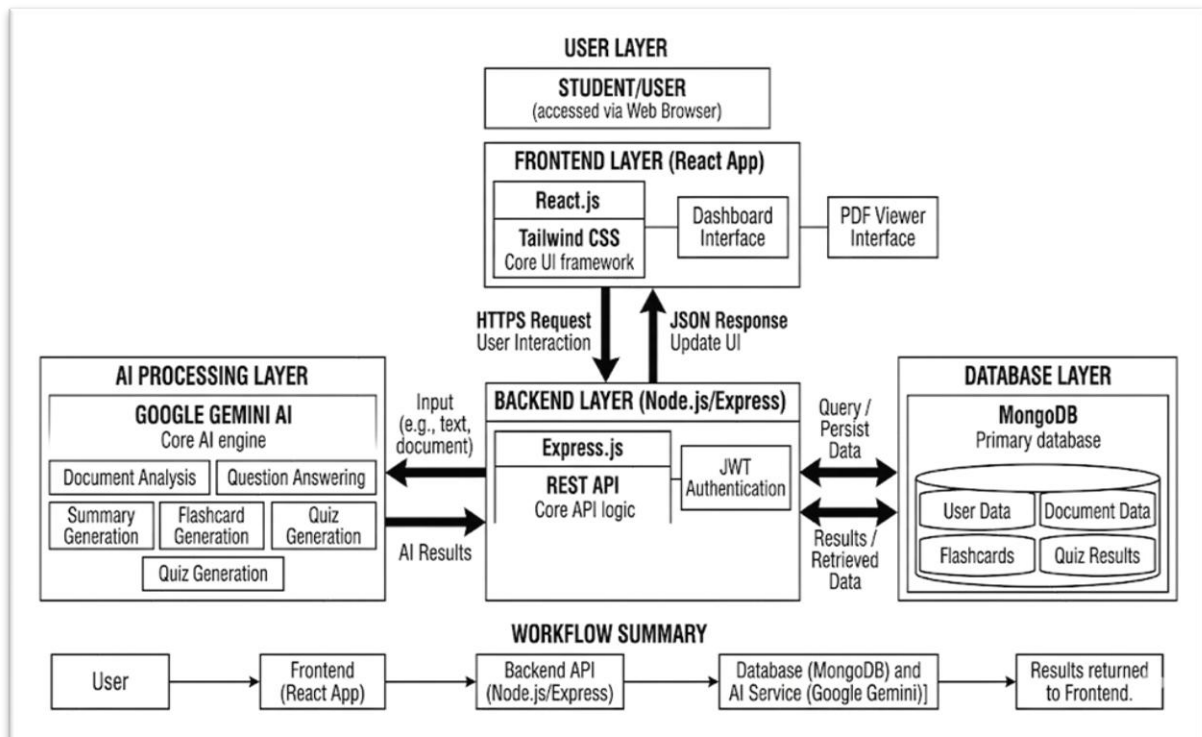


Fig: 3.2

3.3) System Flowchart Diagram:

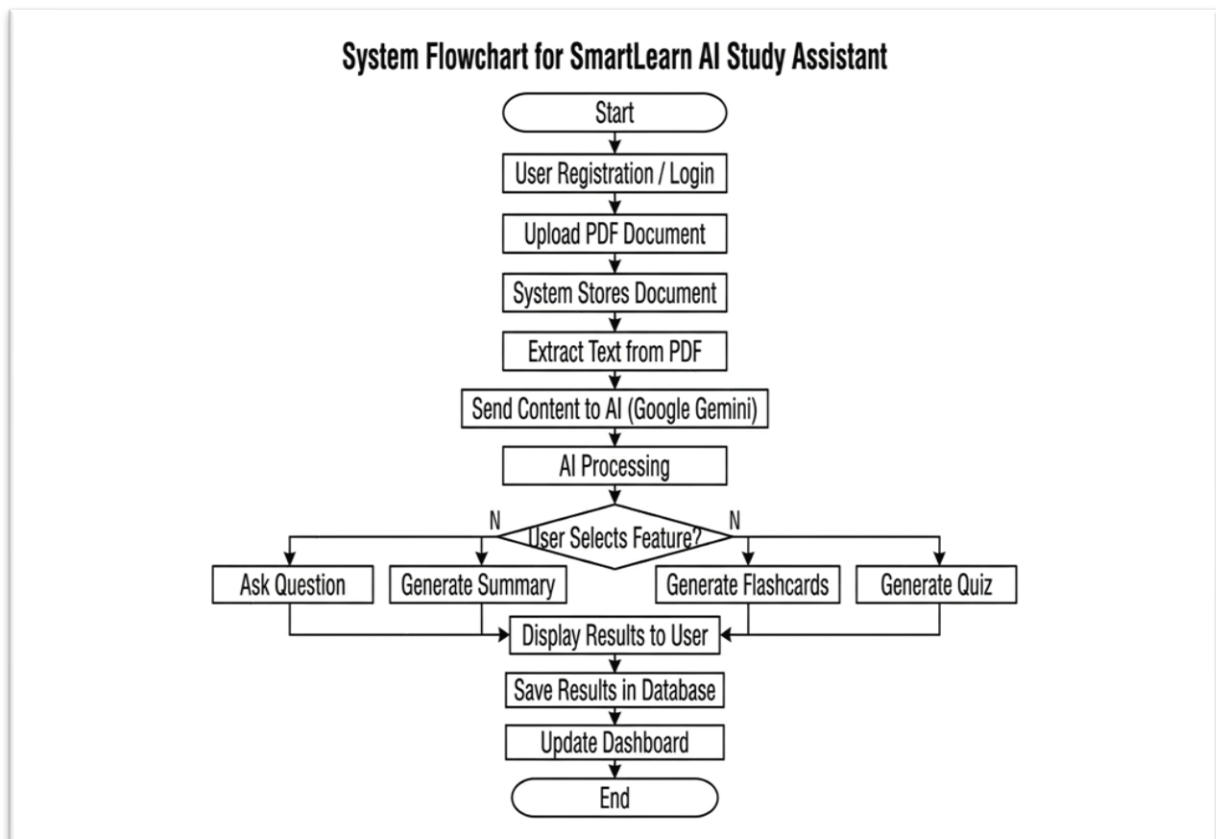


Fig: 3.3

3.4) UML Class Diagram:

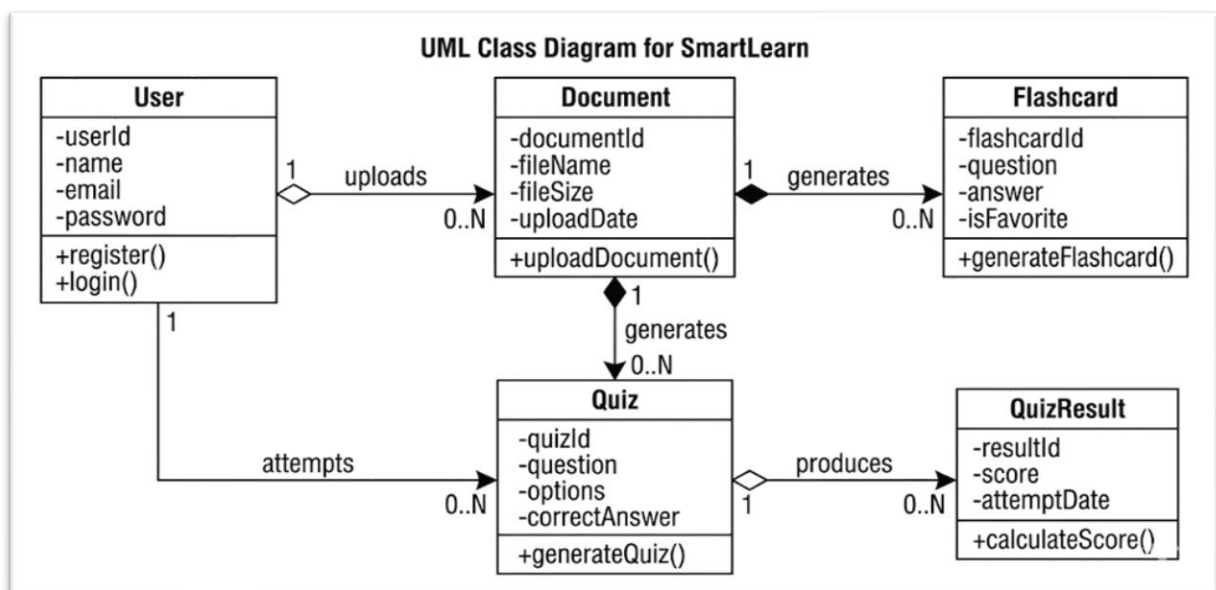


Fig: 3.4

4. Implementation and Results

The SmartLearn platform was successfully implemented using modern web development technologies.

4.1 Implementation

The system includes several modules:

- **User Authentication:** Secure login and registration are implemented using JWT authentication.
- **PDF Upload and Management:** Users can upload PDF documents and manage their study materials.
- **Embedded PDF Viewer:** The system allows users to read uploaded documents directly within the application.
- **AI Chat System:** Users can ask questions related to document content and receive AI-generated responses.
- **Document Summary Generator:** The AI system generates concise summaries of uploaded documents.
- **Flashcard Generator:** Flashcards are automatically created from document content to help students revise quickly.
- **Quiz Generator:** The system generates multiple-choice quizzes based on document information.
- **Dashboard:** The dashboard displays statistics such as total documents, flashcards, quizzes, and recent activities.

4.2 Results

The system successfully demonstrates the integration of artificial intelligence with digital learning tools.

The platform allows students to interact with their study materials in a more efficient way. Screenshots of the working system include the login page, dashboard, document upload interface, AI chat feature, flashcard module, and quiz results page.

These results confirm that SmartLearn provides a functional and intelligent learning environment.

5. Coding

5.1 db.js

```
import mongoose from "mongoose"
import dns from "dns"

dns.setServers([
  '1.1.1.1',
  '8.8.8.8'
])

const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGO_URI);
    console.log(`MongoDB Connected: ${conn.connection.host} ✅`);
  } catch (err) {
    console.error(`Error: ${err.message} ❌`);
    process.exit(1);
  }
};

export default connectDB;
```

5.2 aiRoutes.js

```
import express from "express";
import {
  generateFlashcards,
  generateQuiz,
  generateSummary,
  chat,
  explainConcept,
  getChatHistory,
} from "../controllers/aiController.js";
import protect from "../middleware/auth.js";

const router = express.Router();
router.use(protect);

router.post("/generate-flashcards", generateFlashcards);
router.post("/generate-quiz", generateQuiz);
router.post("/generate-summary", generateSummary);
router.post("/chat", chat);
router.post("/explain-concept", explainConcept);
router.get("/chat-history/:documentId", getChatHistory);

export default router;
```

5.3 authRoutes.js

```
import express from "express";
import { body } from "express-validator";
import {
  register,
  login,
  getProfile,
  updateProfile,
  changePassword,
} from "../controllers/authController.js";
import protect from "../middleware/auth.js";

const router = express.Router();

// Validation middleware
const registerValidation = [
  body("username")
    .trim()
    .isLength({ min: 3 })
    .withMessage("Username must be at least 3 characters long"),
  body("email")
    .isEmail()
    .normalizeEmail()
    .withMessage("Please provide a valid email address"),
  body("password")
    .isLength({ min: 6 })
    .withMessage("Password must be at least 6 characters long"),
];

const loginValidation = [
  body("email")
    .isEmail()
    .normalizeEmail()
    .withMessage("Please provide a valid email address"),
  body("password").notEmpty().withMessage("Password required"),
];

// Public routes
router.post("/register", registerValidation, register);
router.post("/login", loginValidation, login);

// Protected routes
router.get("/profile", protect, getProfile);
router.put("/profile", protect, updateProfile);
router.post("/change-password", protect, changePassword);

export default router;
```

5.4 user.js

```
import mongoose from "mongoose";
import bcrypt from "bcryptjs";

const userSchema = new mongoose.Schema(
  {
    username: {
      type: String,
      required: [true, "Please provide a username"],
      unique: true,
      trim: true,
      minlength: [3, "Username must be at least 3 characters long"],
    },
    email: {
      type: String,
      required: [true, "Please provide an email"],
      unique: true,
      lowercase: true,
      trim: true,
      match: [/^\S+@\S+\.\S+$/, "Please provide a valid email address"],
    },
    password: {
      type: String,
      required: [true, "Please provide a password"],
      minlength: [6, "Password must be at least 6 characters long"],
      select: false,
    },
    profileImage: {
      type: String,
      default: null,
    },
  },
  {
    timestamps: true,
  }
);

// Hash password before saving
userSchema.pre("save", async function(next) {
  if (!this.isModified("password")) {
    next();
  }
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
});

// Compare password method
userSchema.methods.matchPassword = async function (enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};

const User = mongoose.model("User", userSchema);

export default User;
```

7. Prototype Photographs for User Side

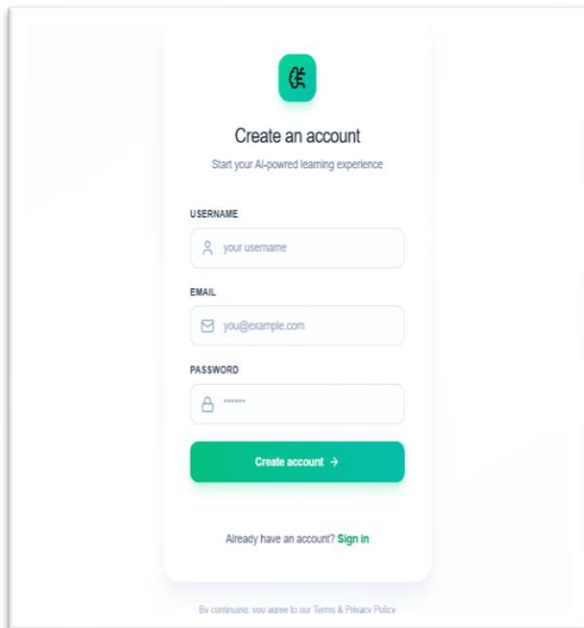


Fig: 7.1

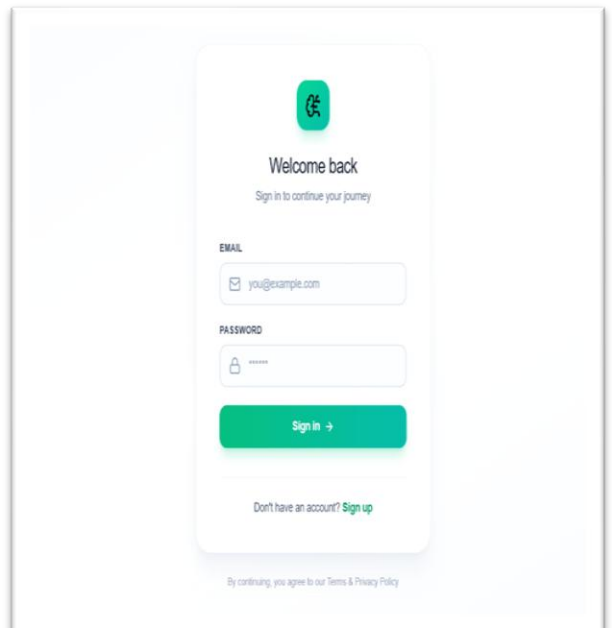


Fig: 7.2

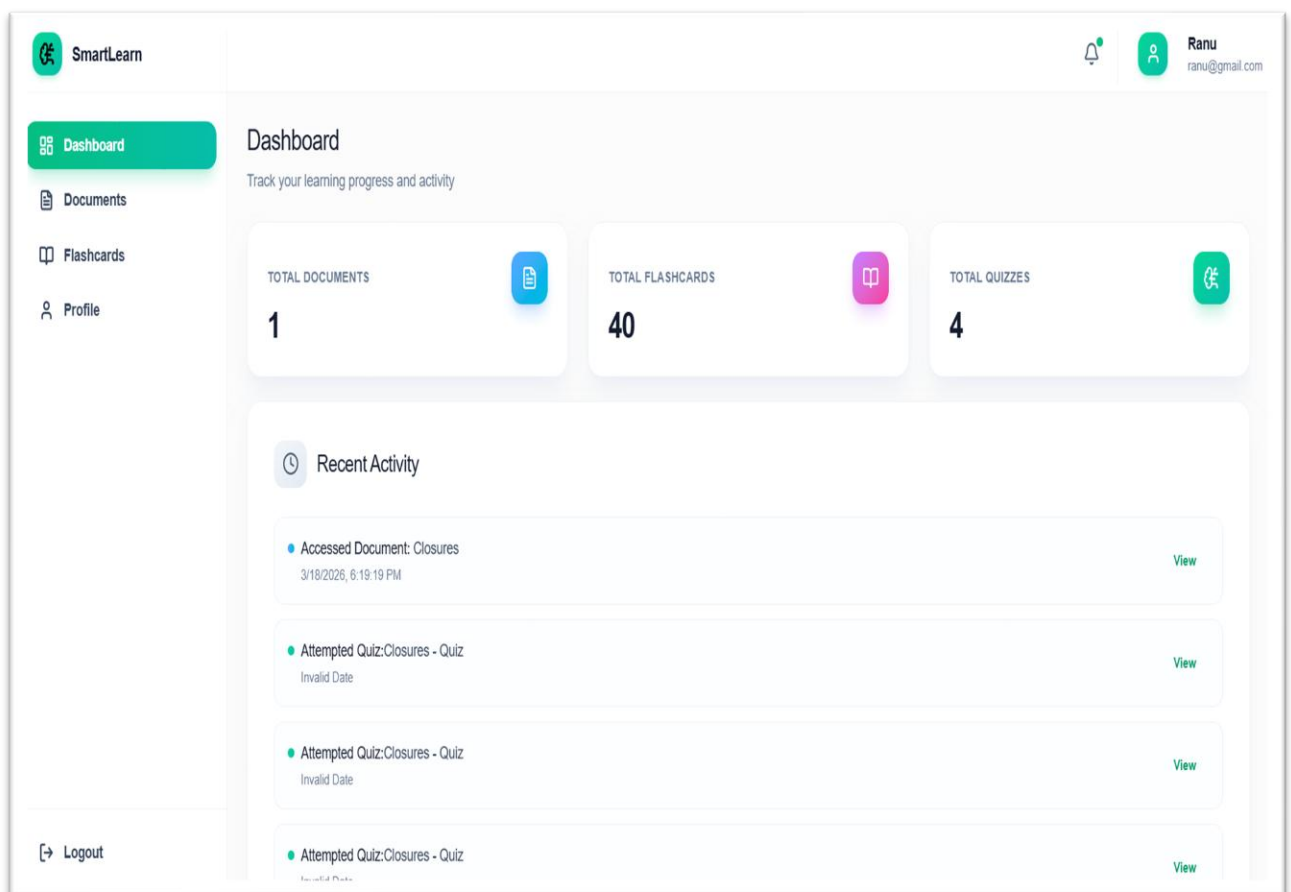


Fig: 7.3

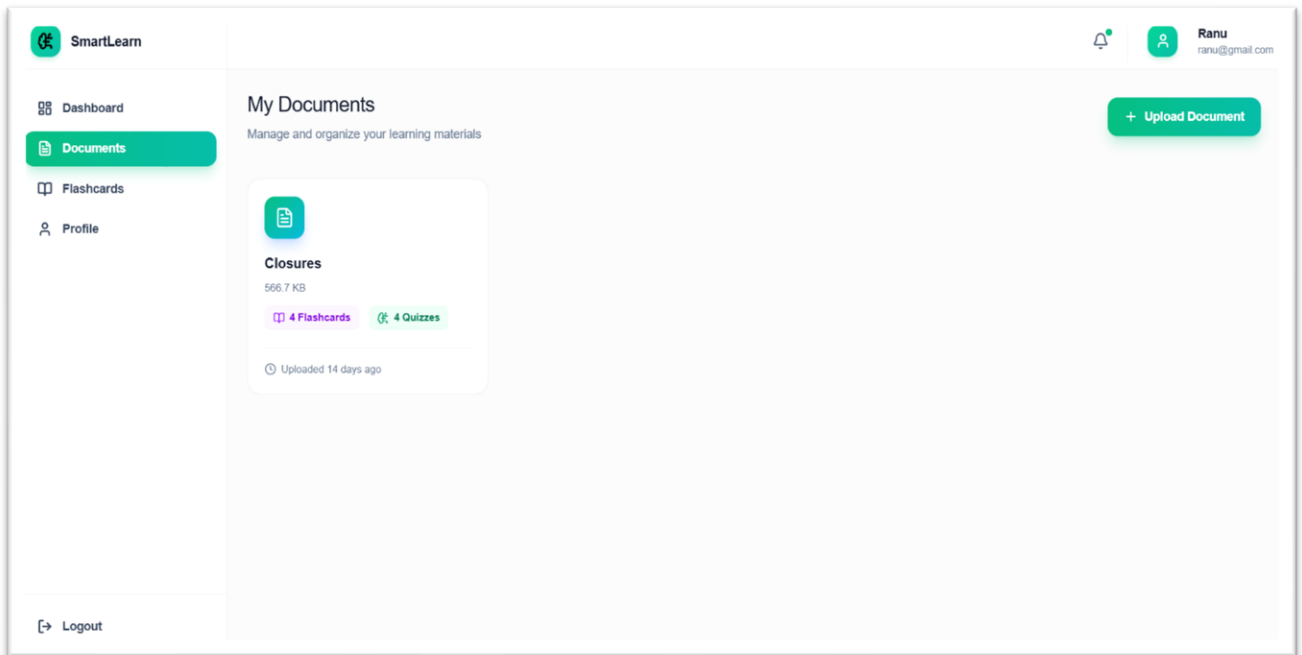


Fig: 7.4

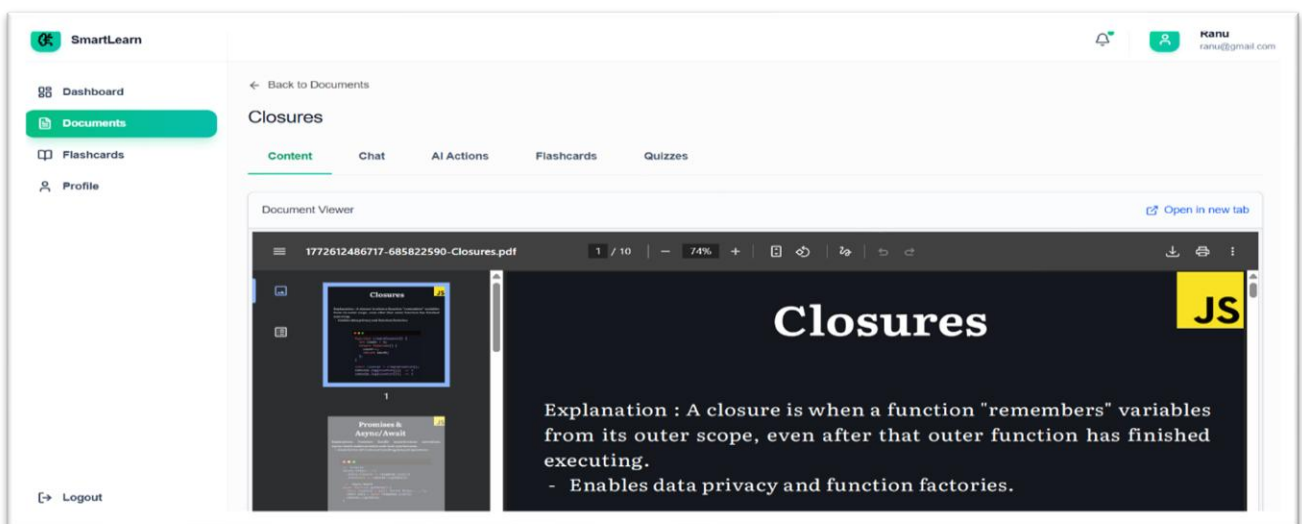


Fig: 7.5

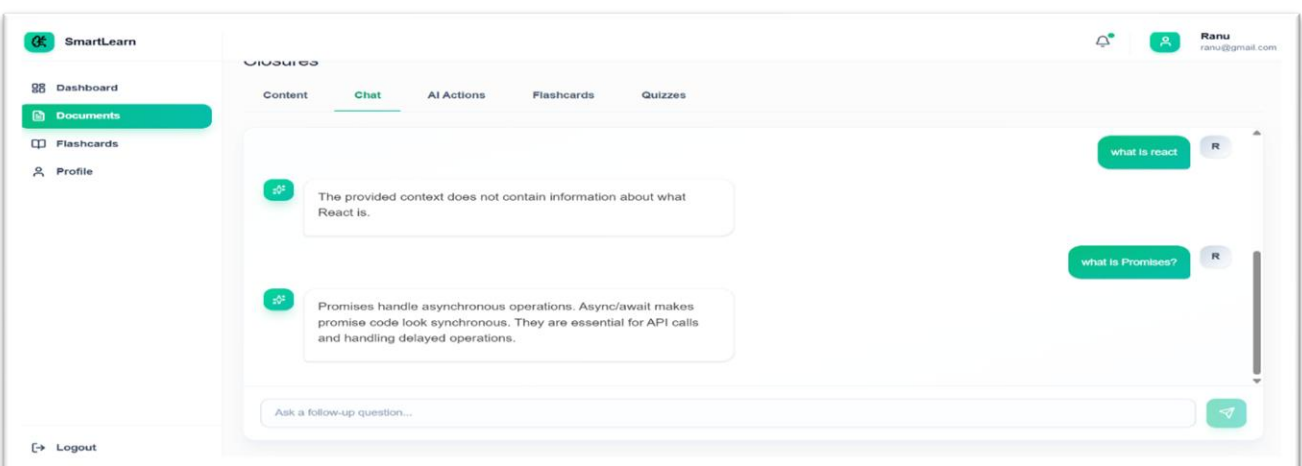


Fig: 7.6

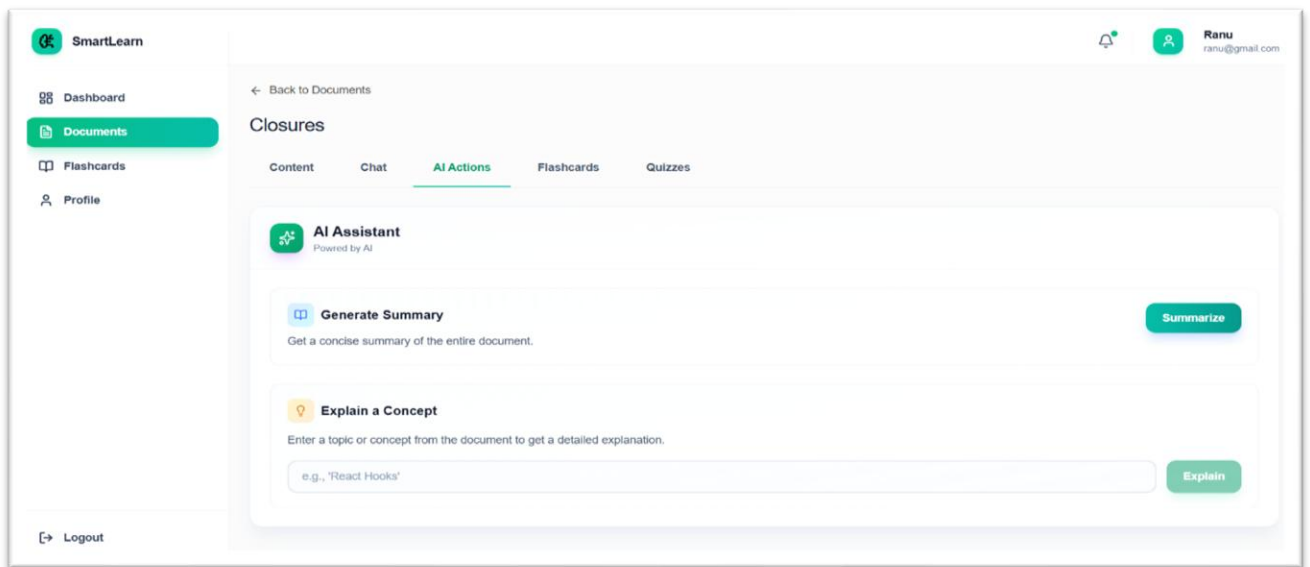


Fig: 7.7

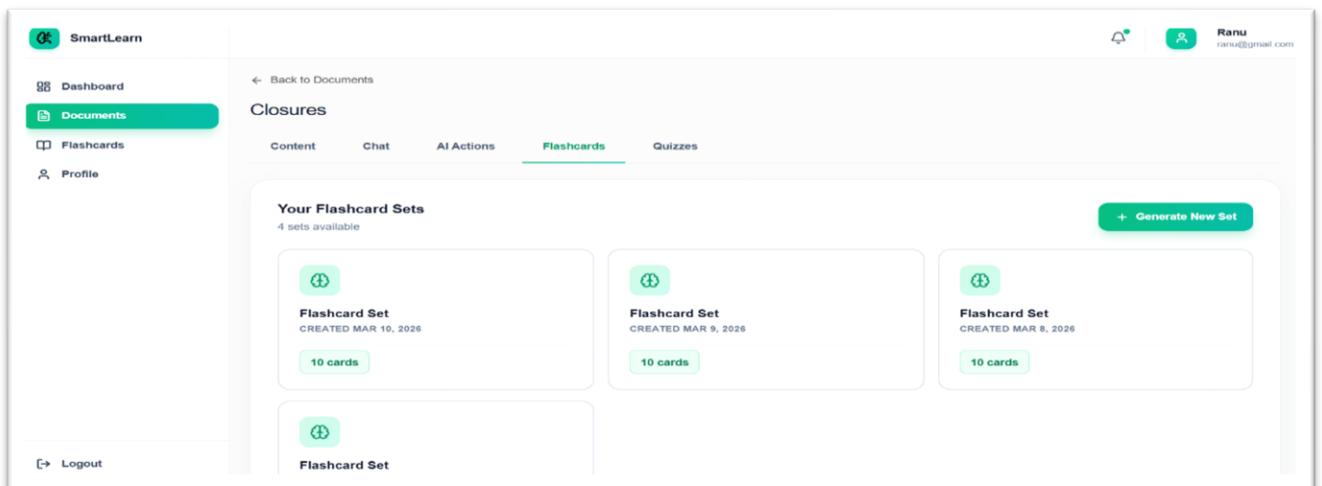


Fig: 7.8

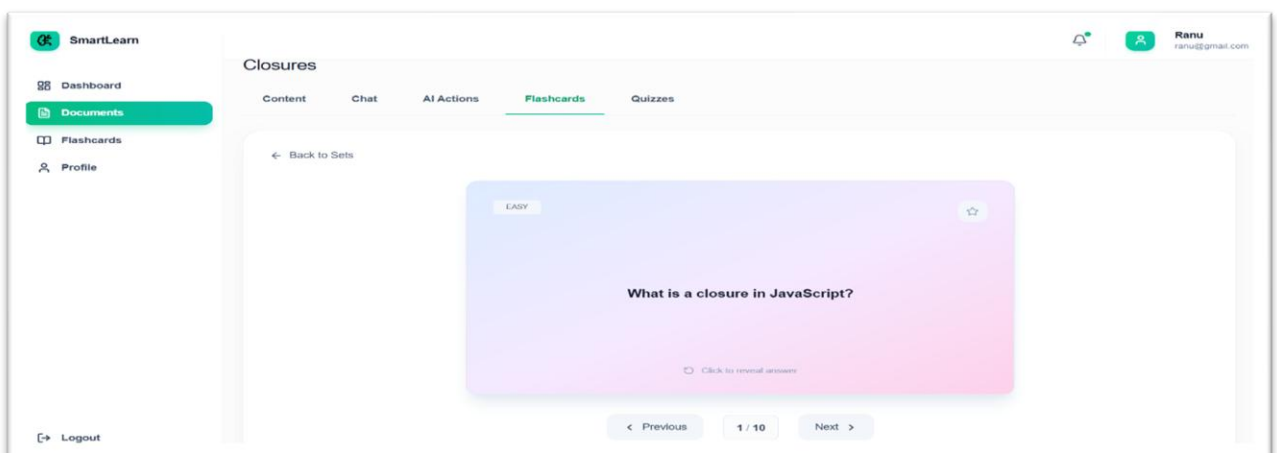


Fig: 7.9

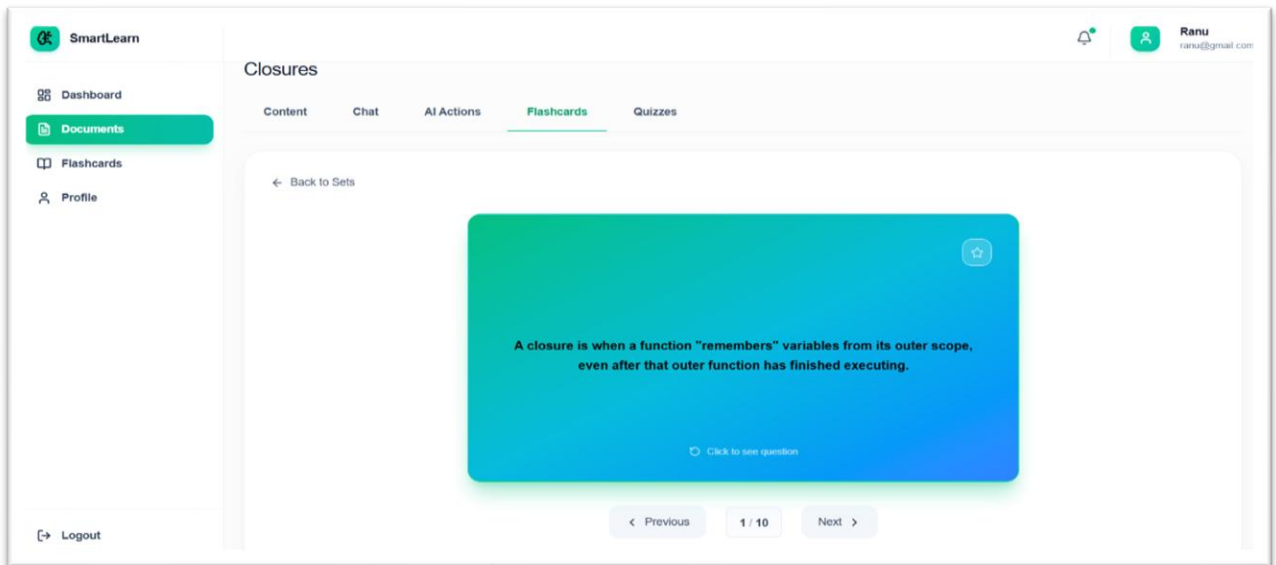


Fig: 7.10

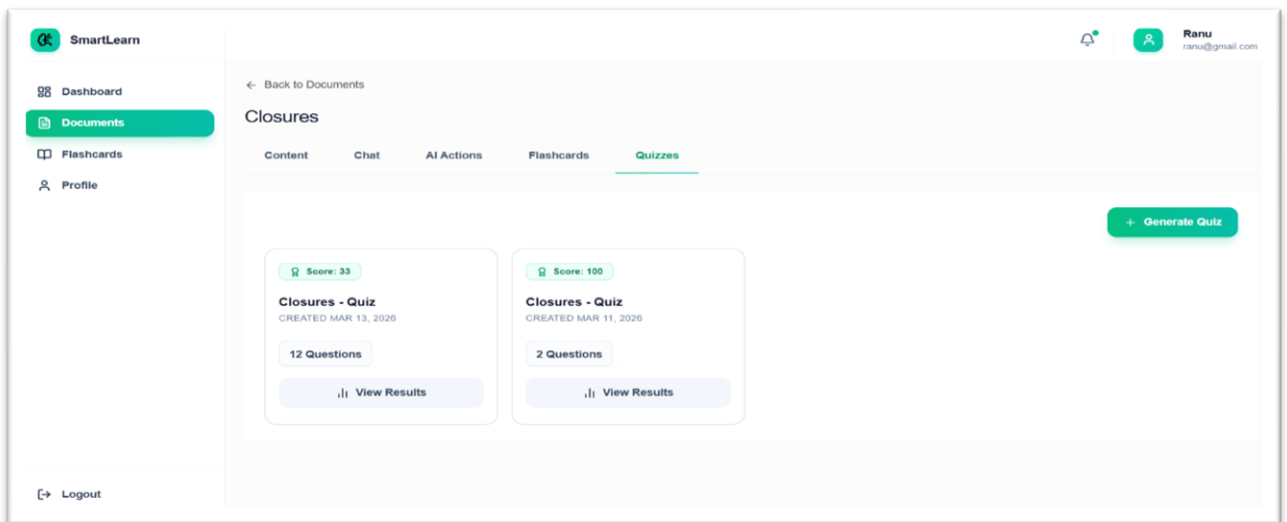


Fig: 7.11

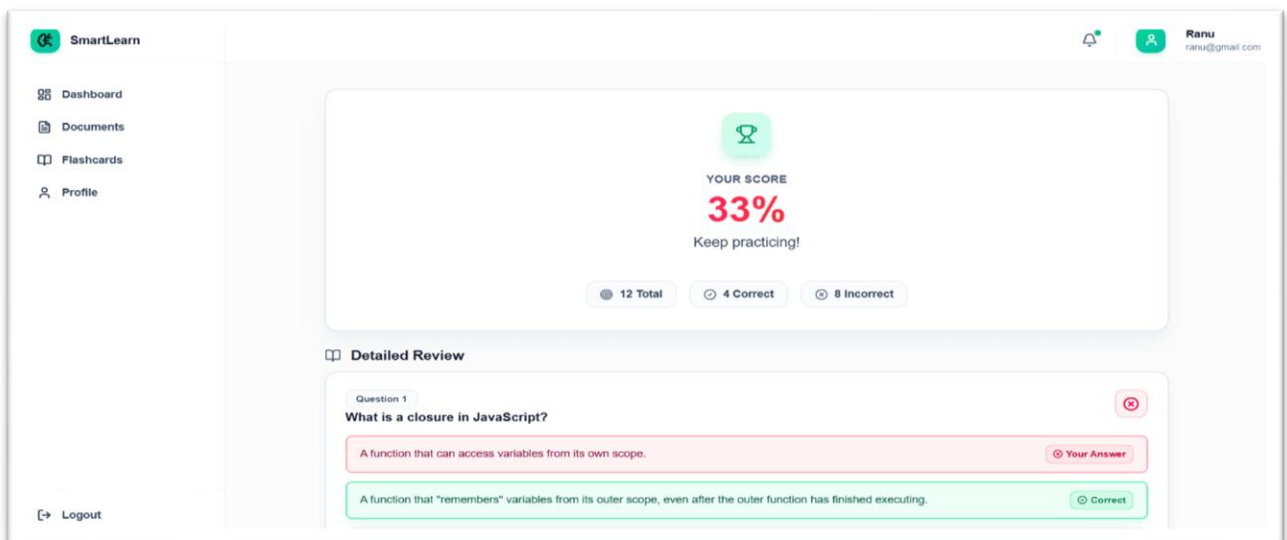


Fig: 7.12

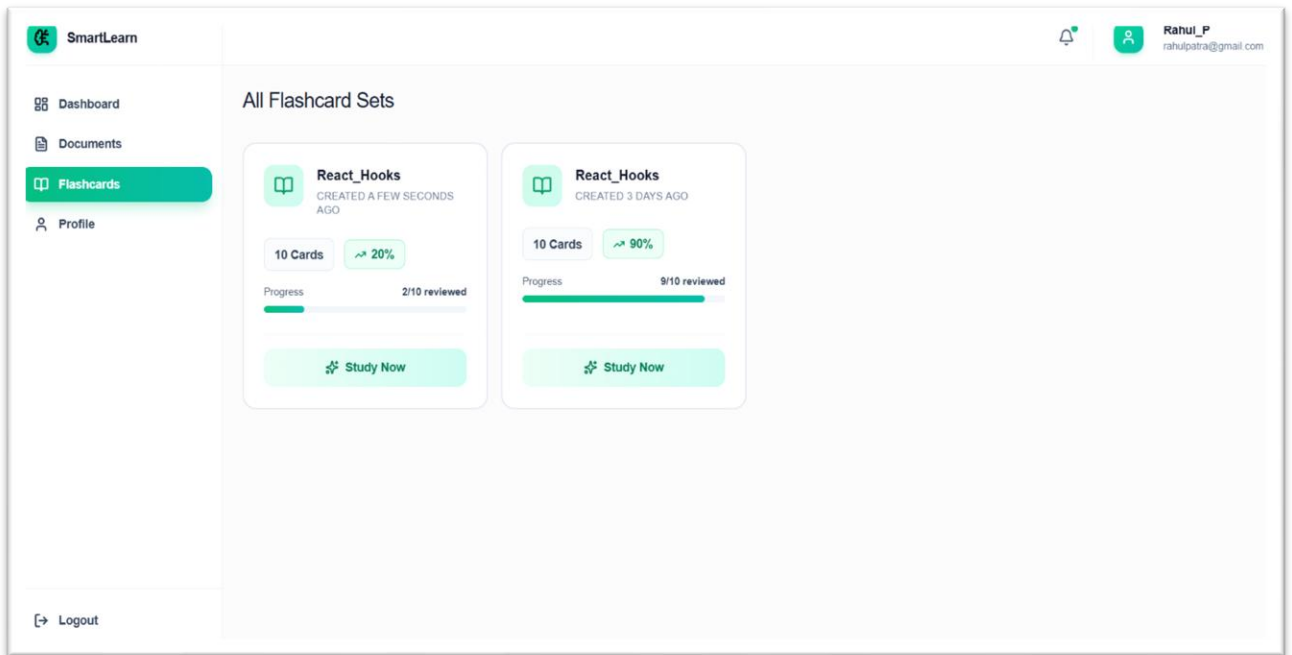


Fig: 7.13

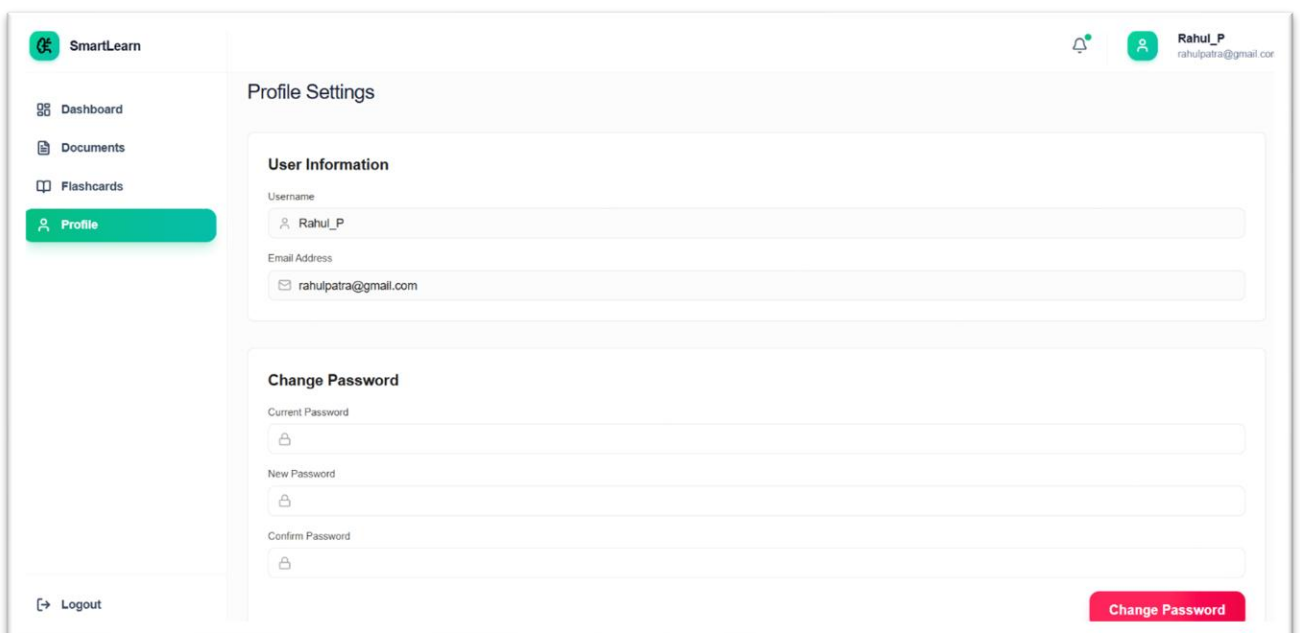


Fig: 7.14

8. Conclusion and Future Work

8.1 Conclusion

The **SmartLearn** project successfully demonstrates the use of artificial intelligence in academic learning systems. By integrating document management with AI-powered features such as summaries, flashcards, quizzes, and question answering, the system provides a more interactive learning environment for students.

The platform helps students reduce the time required to study long documents and supports efficient revision. The combination of modern web technologies and AI services enables SmartLearn to provide intelligent academic support.

8.2 Future Work

Several improvements can be implemented in future versions of the system.

Possible future enhancements include:

- Support for additional document formats such as Word and PowerPoint.
- Advanced analytics for tracking student learning performance.
- Personalized learning recommendations based on user activity.
- Development of a mobile application for improved accessibility.
- Integration of voice-based AI interaction.

These improvements can further enhance the effectiveness of the SmartLearn platform.

9. Bibliography and References

9.1 Websites

1. <https://react.dev/> – Official React documentation for building user interfaces.
2. <https://nodejs.org/en/docs> – Official Node.js documentation for backend development.
3. <https://expressjs.com> – Express.js framework documentation for building web APIs.
4. <https://www.mongodb.com/docs> – MongoDB documentation for database management.

5. <https://tailwindcss.com/docs> – Tailwind CSS documentation for responsive UI design.
6. <https://ai.google.dev/> – Google Gemini AI documentation for implementing AI features.
7. <https://developer.mozilla.org> – MDN Web Docs for JavaScript and web development references.
8. <https://vercel.com/docs> – Vercel deployment documentation for hosting web applications.

9.2 Books

- Learning React by Kirupa Chinnathambi, Addison-Wesley (2nd Edition) – Provides guidance on building modern web applications using React.
- JavaScript: The Good Parts by Douglas Crockford, O'Reilly Media – Explains important concepts of JavaScript used in frontend and backend development.
- Web Development with Node and Express by Ethan Brown, O'Reilly Media – Describes backend development using Node.js and Express.

9.3 Journals

- “Artificial Intelligence in Education: Applications and Future Prospects,” *IEEE Access*, 2023.
- “AI-Based Intelligent Learning Systems for Digital Education,” *International Journal of Educational Technology in Higher Education*, 2022.
- “Natural Language Processing Techniques for Document Understanding,” *IEEE Transactions on Knowledge and Data Engineering*, 2021.